



Study Questionnaire

Creating Novel Virtual Reality Interfaces for Teaching Computer Architecture

Participant ID: _____

Date: _____

Section A: Screening and Background

1. Are you 18 or older?

☐ Yes ☐ No

2. Have you previously studied computer architecture or computer organisation?

☐ Yes ☐ No

3. How familiar are you with virtual reality?

☐ None ☐ Limited ☐ Moderate ☐ Good ☐ High

4. Have you used a VR headset before?

☐ Yes ☐ No

5. Before this session, how confident are you in your understanding of computer architecture concepts?

☐ Strongly disagree ☐ Disagree ☐ Neither agree nor disagree ☐ Agree ☐ Strongly agree

6. How difficult do you usually find topics such as instruction flow, pipelining, stack behaviour, and memory hierarchy?

☐ Very easy ☐ Easy ☐ Neutral ☐ Difficult ☐ Very difficult

7. Which of the following topics have you previously studied? Tick all that apply.

☐ Instruction flow ☐ Pipelining ☐ Stack ☐ Memory hierarchy ☐ Assembly
☐ None

Section B: System Usability

For Questions 8–14, please select one response for each statement.

8. The VR system was easy to learn.

☐ Strongly disagree ☐ Disagree ☐ Neither agree nor disagree ☐ Agree ☐
Strongly agree

9. I could navigate the environment without difficulty.

☐ Strongly disagree ☐ Disagree ☐ Neither agree nor disagree ☐ Agree ☐
Strongly agree

10. The controls felt intuitive.

☐ Strongly disagree ☐ Disagree ☐ Neither agree nor disagree ☐ Agree ☐
Strongly agree

11. The visual presentation made the concepts easier to follow.

☐ Strongly disagree ☐ Disagree ☐ Neither agree nor disagree ☐ Agree ☐
Strongly agree

12. I could focus on the learning task without being distracted by the interface.

☐ Strongly disagree ☐ Disagree ☐ Neither agree nor disagree ☐ Agree ☐
Strongly agree

13. I would feel confident using this system again.

☐ Strongly disagree ☐ Disagree ☐ Neither agree nor disagree ☐ Agree ☐
Strongly agree

14. I experienced discomfort, motion sickness, or visual fatigue while using the system.

☐ Strongly disagree ☐ Disagree ☐ Neither agree nor disagree ☐ Agree ☐
Strongly agree

Section C: System Engagement

For Questions 15–19, please select one response for each statement.

15. The experience held my attention.

☐ Strongly disagree ☐ Disagree ☐ Neither agree nor disagree ☐ Agree ☐
Strongly agree

16. I felt actively involved in the learning process.

☐ Strongly disagree ☐ Disagree ☐ Neither agree nor disagree ☐ Agree ☐
Strongly agree

17. The system made the topic feel more interesting.

☐ Strongly disagree ☐ Disagree ☐ Neither agree nor disagree ☐ Agree ☐
Strongly agree

18. I was motivated to explore the concepts further.

☐ Strongly disagree ☐ Disagree ☐ Neither agree nor disagree ☐ Agree ☐
Strongly agree

19. The interactive format improved my focus compared with standard learning materials.

☐ Strongly disagree ☐ Disagree ☐ Neither agree nor disagree ☐ Agree ☐
Strongly agree

Section D: Perceived Learning

For Questions 20–24, please select one response for each statement.

20. The VR experience helped me understand the topic better.

☐ Strongly disagree ☐ Disagree ☐ Neither agree nor disagree ☐ Agree ☐
Strongly agree

21. The visualisations helped me build a mental model of what was happening.

☐ Strongly disagree ☐ Disagree ☐ Neither agree nor disagree ☐ Agree ☐
Strongly agree

22. The system made abstract processes easier to understand.

☐ Strongly disagree ☐ Disagree ☐ Neither agree nor disagree ☐ Agree ☐ Strongly agree

23. I would prefer this as a supplement to traditional teaching materials.

☐ Strongly disagree ☐ Disagree ☐ Neither agree nor disagree ☐ Agree ☐ Strongly agree

24. I believe I learned something useful from this session.

☐ Strongly disagree ☐ Disagree ☐ Neither agree nor disagree ☐ Agree ☐ Strongly agree

Section E: Conceptual Questions

For each question, tick one answer. A confidence rating can be included if you want extra analysis, but I recommend keeping it optional rather than mandatory unless confidence is a core study outcome.

25. Which register stores the address of the next instruction?

☐ Stack Pointer ☐ Program Counter ☐ Instruction Register ☐ Base Register

Optional confidence rating: ☐ 1 ☐ 2 ☐ 3 ☐ 4 ☐ 5

26. Which stage of the instruction cycle reads an instruction from memory?

☐ Decode ☐ Fetch ☐ Execute ☐ Write-back

Optional confidence rating: ☐ 1 ☐ 2 ☐ 3 ☐ 4 ☐ 5

27. During instruction execution, the instruction register primarily stores:

☐ Program variables ☐ The currently executing instruction ☐ The stack pointer value ☐ Cache addresses

Optional confidence rating: ☐ 1 ☐ 2 ☐ 3 ☐ 4 ☐ 5

28. What happens to the program counter after an instruction is fetched?

☐ It resets to zero ☐ It increments to the next instruction address ☐ It copies into the stack pointer ☐ It stores the result of the ALU

Optional confidence rating: ☐ 1 ☐ 2 ☐ 3 ☐ 4 ☐ 5

29. What is the main purpose of the stack during function calls?

☐ Store pipeline stages ☐ Store return addresses and local variables ☐ Store instruction cache data ☐ Store CPU clock values

Optional confidence rating: ☐ 1 ☐ 2 ☐ 3 ☐ 4 ☐ 5

30. Which register tracks the top of the stack?

☐ PC ☐ SP ☐ IR ☐ ALU

Optional confidence rating: ☐ 1 ☐ 2 ☐ 3 ☐ 4 ☐ 5

31. When a function returns, which of the following usually happens?

☐ Stack pointer moves back to previous position ☐ Program counter resets to zero
☐ Cache is cleared ☐ Registers are deleted

Optional confidence rating: ☐ 1 ☐ 2 ☐ 3 ☐ 4 ☐ 5

32. Which event normally pushes data onto the stack?

☐ Memory read ☐ Function call ☐ Pipeline stall ☐ Cache miss

Optional confidence rating: ☐ 1 ☐ 2 ☐ 3 ☐ 4 ☐ 5

33. Why do CPUs use pipelining?

☐ To reduce instruction memory ☐ To increase instruction throughput ☐ To eliminate branching ☐ To reduce stack usage

Optional confidence rating: ☐ 1 ☐ 2 ☐ 3 ☐ 4 ☐ 5

34. Which of the following best describes a pipeline stall?

☐ A new instruction enters the pipeline ☐ The pipeline pauses due to a dependency or hazard ☐ The stack pointer changes ☐ The ALU executes multiple instructions

Optional confidence rating: ☐ 1 ☐ 2 ☐ 3 ☐ 4 ☐ 5

35. A data hazard occurs when:

☐ An instruction depends on the result of a previous instruction ☐ Memory latency increases ☐ Branch prediction fails ☐ Cache memory overflows

Optional confidence rating: ☐ 1 ☐ 2 ☐ 3 ☐ 4 ☐ 5

36. Which pipeline stage usually performs arithmetic operations?

☐ Fetch ☐ Decode ☐ Execute ☐ Write-back

Optional confidence rating: ☐ 1 ☐ 2 ☐ 3 ☐ 4 ☐ 5

37. Branch instructions primarily affect which component?

☐ Cache memory ☐ Program counter ☐ Stack pointer ☐ Instruction register

Optional confidence rating: ☐ 1 ☐ 2 ☐ 3 ☐ 4 ☐ 5

38. If a branch prediction is incorrect:

☐ The CPU must flush incorrect instructions in the pipeline ☐ The stack pointer resets ☐ The instruction register deletes memory ☐ The ALU halts

Optional confidence rating: ☐ 1 ☐ 2 ☐ 3 ☐ 4 ☐ 5

39. Why are caches used in modern CPUs?

☐ To increase instruction size ☐ To store program code permanently ☐ To reduce average memory access time ☐ To remove the need for RAM

Optional confidence rating: ☐ 1 ☐ 2 ☐ 3 ☐ 4 ☐ 5

40. Which memory type is fastest?

☐ Hard disk ☐ RAM ☐ Cache ☐ Virtual memory

Optional confidence rating: ☐ 1 ☐ 2 ☐ 3 ☐ 4 ☐ 5

41. Which memory component is typically largest but slowest?

☐ Cache ☐ Registers ☐ Main memory ☐ Secondary storage

Optional confidence rating: ☐ 1 ☐ 2 ☐ 3 ☐ 4 ☐ 5

42. What does this instruction do? `add x1, x2, x3`

☐ Adds `x2` and `x3` and stores the result in `x1`
☐ Adds `x1` and `x2` and stores the result in `x3`
☐ Moves `x2` into `x3`
☐ Multiplies registers

Optional confidence rating: ☐ 1 ☐ 2 ☐ 3 ☐ 4 ☐ 5

43. What does this instruction do? `addi x5, x0, 10`

☐ Stores 10 into `x5` ☐ Clears `x5` ☐ Moves `x5` into `x0` ☐ Adds two registers

Optional confidence rating: ☐ 1 ☐ 2 ☐ 3 ☐ 4 ☐ 5

44. Which instruction type loads data from memory into a register?

☐ ADD ☐ STORE ☐ LOAD ☐ JUMP

Optional confidence rating: ☐ 1 ☐ 2 ☐ 3 ☐ 4 ☐ 5

45. Consider this sequence: `lw x1, 0(x2)` followed by `add x3, x1, x4`. Why might a pipeline stall occur?

☐ Because the second instruction depends on `x1`
☐ Because `x2` is unused
☐ Because the program counter resets
☐ Because memory is cleared

Optional confidence rating: ☐ 1 ☐ 2 ☐ 3 ☐ 4 ☐ 5

46. Which register usually holds the return address after a function call in RISC-V?

☐ x0 ☐ ra ☐ sp ☐ gp

Optional confidence rating: ☐ 1 ☐ 2 ☐ 3 ☐ 4 ☐ 5

47. What is the role of register x0 in RISC-V?

☐ Stores the program counter ☐ Always contains the value 0 ☐ Stores return addresses ☐ Tracks the stack pointer

Optional confidence rating: ☐ 1 ☐ 2 ☐ 3 ☐ 4 ☐ 5

48. What does this instruction do? lw x5, 0(x10)

☐ Stores x5 into memory ☐ Loads a word from memory into x5 ☐ Moves x10 into x5 ☐ Clears memory

Optional confidence rating: ☐ 1 ☐ 2 ☐ 3 ☐ 4 ☐ 5

49. What does this instruction do? sw x5, 4(x10)

☐ Value from x5 is stored in memory ☐ Value from memory is loaded into x5 ☐ x10 is overwritten ☐ Stack pointer moves

Optional confidence rating: ☐ 1 ☐ 2 ☐ 3 ☐ 4 ☐ 5

50. What is this instruction commonly used for? addi sp, sp, -16

☐ Allocate space on the stack ☐ Reset stack pointer ☐ Jump to a function ☐ Clear registers

Optional confidence rating: ☐ 1 ☐ 2 ☐ 3 ☐ 4 ☐ 5

51. What is this instruction commonly used for? addi sp, sp, 16

☐ Allocates stack memory ☐ Frees stack memory after a function returns ☐ Loads memory ☐ Changes the program counter

Optional confidence rating: ☐ 1 ☐ 2 ☐ 3 ☐ 4 ☐ 5

52. What does this instruction do? beq x1, x2, label

☐ Jump if x1 > x2 ☐ Jump if x1 == x2 ☐ Jump if x1 < x2 ☐ Always jump

Optional confidence rating: ☐ 1 ☐ 2 ☐ 3 ☐ 4 ☐ 5

53. What does this instruction do? jal ra, func

☐ Calls a function and stores the return address in ra
☐ Jumps without saving the return address
☐ Loads memory
☐ Writes to the stack

Optional confidence rating: ☐ 1 ☐ 2 ☐ 3 ☐ 4 ☐ 5

54. Put the instruction stages in the correct order.

☐ Execute, Fetch, Decode, Write-back

- ☐ Fetch, Decode, Execute, Write-back
- ☐ Decode, Fetch, Execute, Write-back
- ☐ Fetch, Execute, Decode, Write-back

Optional confidence rating: ☐ 1 ☐ 2 ☐ 3 ☐ 4 ☐ 5

Section F: Open-Ended Feedback

55. What part of the experience helped your understanding most? _____

56. What part was confusing or ineffective? _____

57. What would you improve in the system? _____

58. Were there any moments where the VR interface got in the way of learning? _____

59. Would you use this again as a learning aid? Why or why not?
